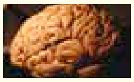
YOUR BRAIN V/S A SUPERCOMPUTER



Have you ever wondered how powerful a brain really is? The number of flops may be accurate, but not too intuitive. Japanese scientists have reported that even the most powerful supercomputer takes nearly 40 minutes to simulate one second of brain activity! http://dgit.in/1juCX9l

THE CASE AGAINST SPOTIFY

An Indie musician talks about how Spotify and other streaming services may not be musician-friendly. While these are great options for listeners, artists get a raw deal. Royalties amount to as low as $0.003 per listen. http://dgit.in/1moBj8M

REDDIT'S FAVOURITE BOOKS

If you're a book lover, you've faced the question of what to read next. Sit back, and let the Internet help you out. Reddit user Raerth has compiled a massive list of the most up voted books from across multiple Reddit threads – over 200 books. http://dgit.in/1anlsXg

SILICON VALLEY - A CLASS DIVIDED

America's Silicon Valley may be the seat of the brightest minds in tech, but it's not without problems. As engineers' salaries sail through the roof, the class divide increases. Protesters surrounded a bus carrying Google employees only serves to highlight this problem. http://dgit.in/Kzp5hY

Smartphone Cameras: 5 ways they can improve in 2014

It's a new year and we're starting it off by hoping that one of the most popular forms of photography will gain some serious technological advancement in the next 363 days.

- By Swapnil Mathur

The previous year was pretty big for cellphone photography. Nokia dropped the 41MP bomb-shell, Sony packed a larger sensor into their water-resistant Xperia Z1. The iPhone improved upon certain aspects of their camera performance too; but 2014 is a new year and we are fresh on hope. Here is a list of five features that we are hoping cellphone and imaging companies will innovate on in order to take cellphone imaging to the next level.

About the optics

While sensors are getting bigger, what we see lacking is optics. Even today, most lens elements in a camera phone are made from plastic. It may be high grade plastic, but it's still not as good as its glass counterpart. One of the reasons glass isn't used in cellphones is because of the rugged conditions a cellphone is subjected to. That was also why a mechanical shutter was never introduced into camera phones, but the Nokia Lumia 1020 defied that. One can only hope that phone manufacturers will figure out a way to add glass to the assembly, without compromising the integrity of the device.

Low-light wizardry

Even though a lot of the modern cellphones produce decent enough photos in low-light (as long as you see them on the phone and nothing bigger), but they aren't *great*. We hope that this year, manufacturers would improve upon the low-light performance by pushing the apertures as far as f/1.4 (or even f/1.0 if we had to be ambitious). However, that would have to be coupled up with a better image processing engine as well.

More juice!

Screens are definitely the most battery draining component on a cellphone, but when you kick that camera into the on position, you're taxing your battery, hard. Switching from photo to video makes it worse. The whole camera thing is such a drain that Nokia bundled a spare 1020 mAh battery into the battery grip accessory. Even with that kind of power, the Lumia 1020 barely lasted the day, with somewhat heavy usage, of course. So with better cameras, we await better batteries, or at least better battery life.

Unleash the RAW power!

One of the drawbacks of camera imaging has been the little control users have over the nature of the output. There's not a lot of scope left around for editing (unless you're just using Instagram), which is a bit of a turn off, especially if you've invested in a *good* camera phone. Nokia fixed this by adding RAW DNG capture support to some of their high end Lumia phones like the Lumia 1520/1020. We'd still like to see the likes of Sony, LG, Samsung and Apple bring RAW capture support to their phones. Using Adobe's open DNG format,would save a lot of R&D.

A useful flash

Many have attempted to make the cellphone camera flash actually usable, with very little success. Nokia threw a curve ball when they packed an uber-powerful Xenon flash tube into the Lumia 1020 which showed promise. While the power was there, the flash failed at delivering correct colour temperature to photos, often causing a green cast. Apple too introduced the "TrueTone" flash with the iPhone 5s, but even that leaves much to be desired. It would be nice to see the cellphones come with a flash with a little more effect in terms of intensity and colour temperature so that the photos are not only well-exposed, but also have a more natural colour tone.

While these are definitely the top 5 innovations we'd like to see in our cellphones in 2014, we wouldn't mind a Lytro-type functionality, either. And since most of these companies are probably paying attention, we wouldn't mind a built-in neutral density filter or a Polarizer either. Kthxbai.